Earth-2 Overview

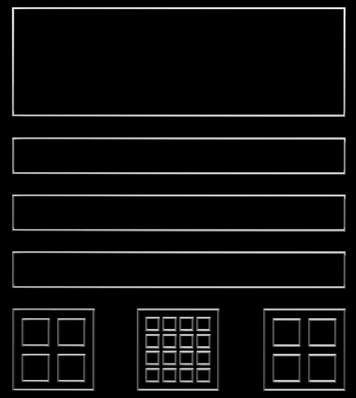
Jay Chen | Solution Architect, Nov 2024.

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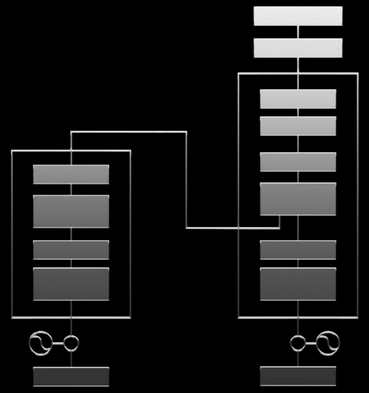
Earth-2 Platform

3 Pillars - HPC & AI & VISUALIZATION

CWA Earth-2 Applications



HPC



AI



VISUALIZATION

Earth-2 Platform

CPU | GPU | DPU

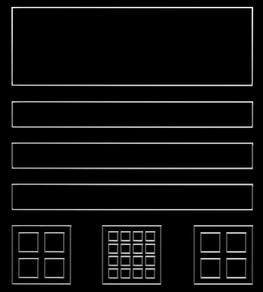
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Earth-2 @ CWA

WRF (CPU)

FV3 (CPU)

TCo (GPU)



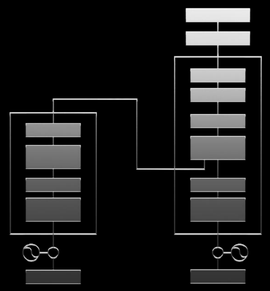
HPC SDK

HPC

FourCastNet-v2

CorrDiff

Coming Soon…



Modulus | Earth2Studio | NIM

AI

ParaView+IndeX

NVIDIA GTC 24

Computex 24



OMNIVERSE

VISUALIZATION

Earth-2 Platform

CPU | GPU | DPU

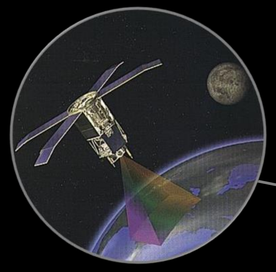
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Earth-2 Platform

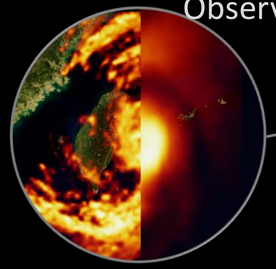
Connecting Complex Simulation, Data and AI Workflows



NWP Simulations



Observations



AI Simulations



3D GIS Data



HPC SDK



MODULUS

SIMULATION | **AI**

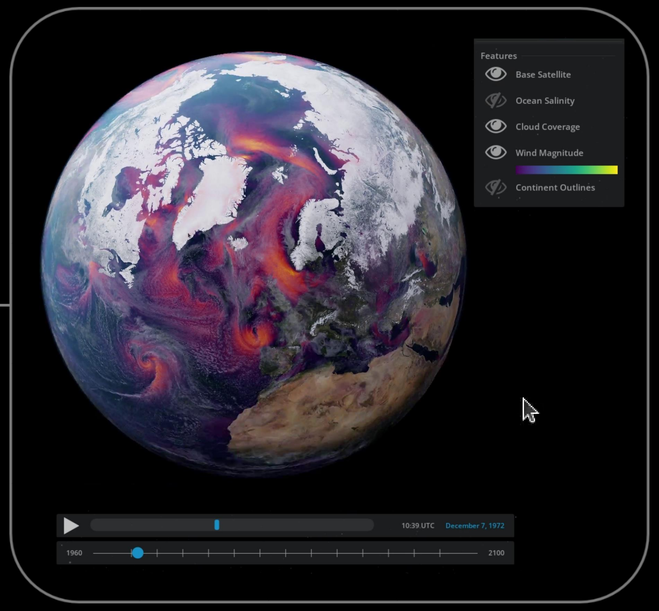
VISUALIZATION



OMNIVERSE



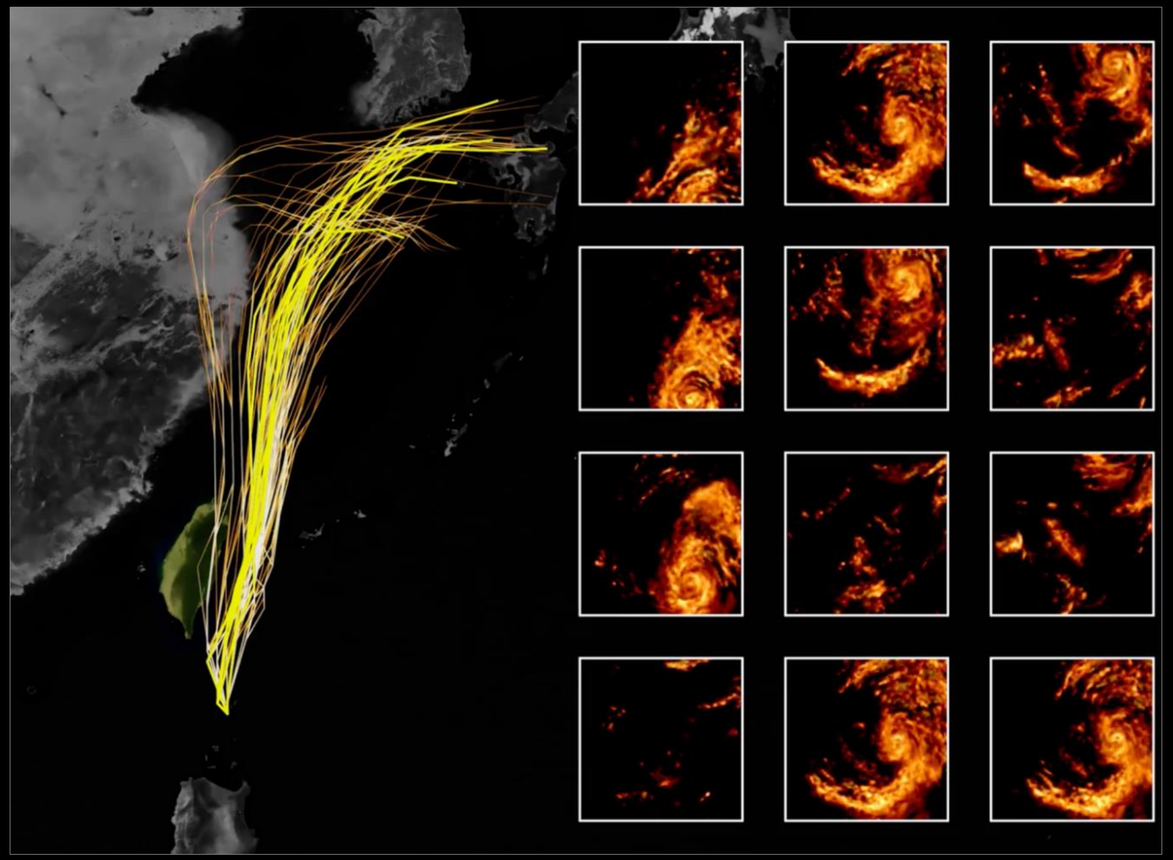
Renderer



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Earth-2 Digital Twin

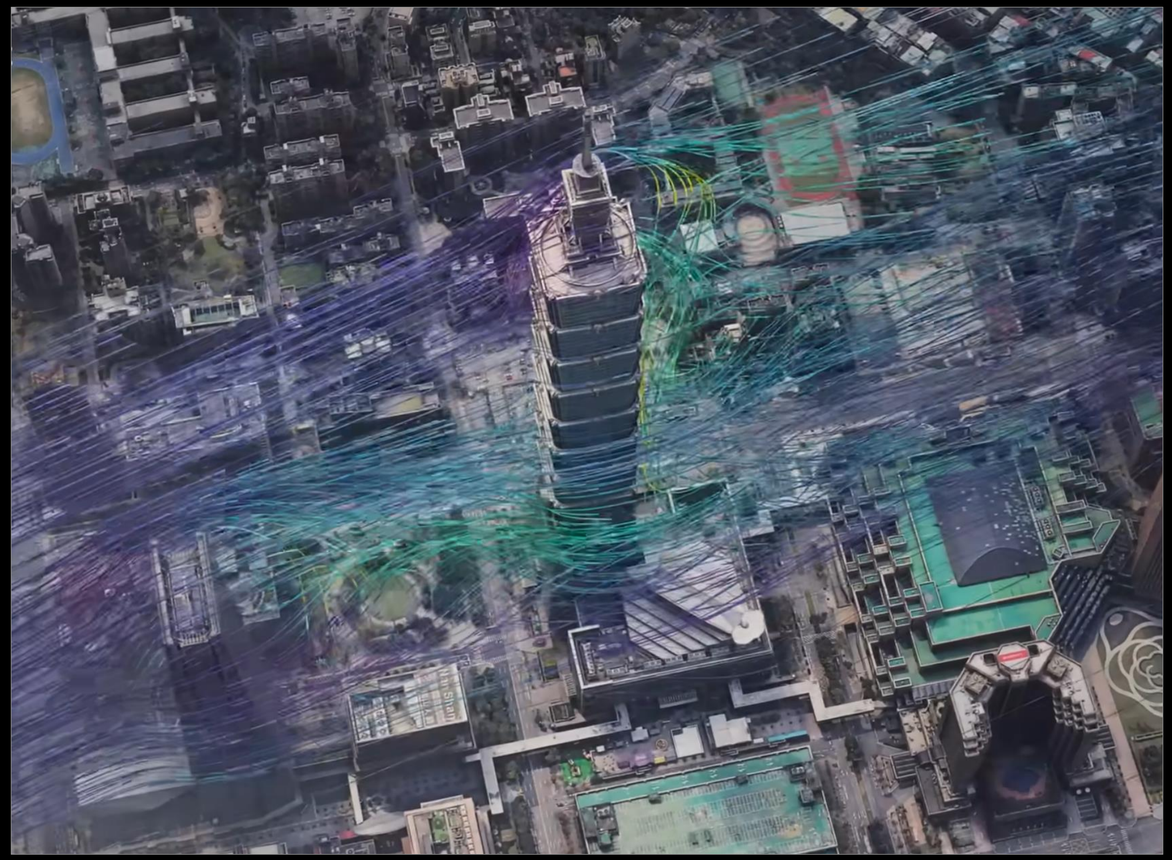
Based on Omniverse



GTC 2024-March

FourCastNet and CorrDiff Ensemble of Typhoon Track Predictions

VISUALIZATION



Computex 2024-June

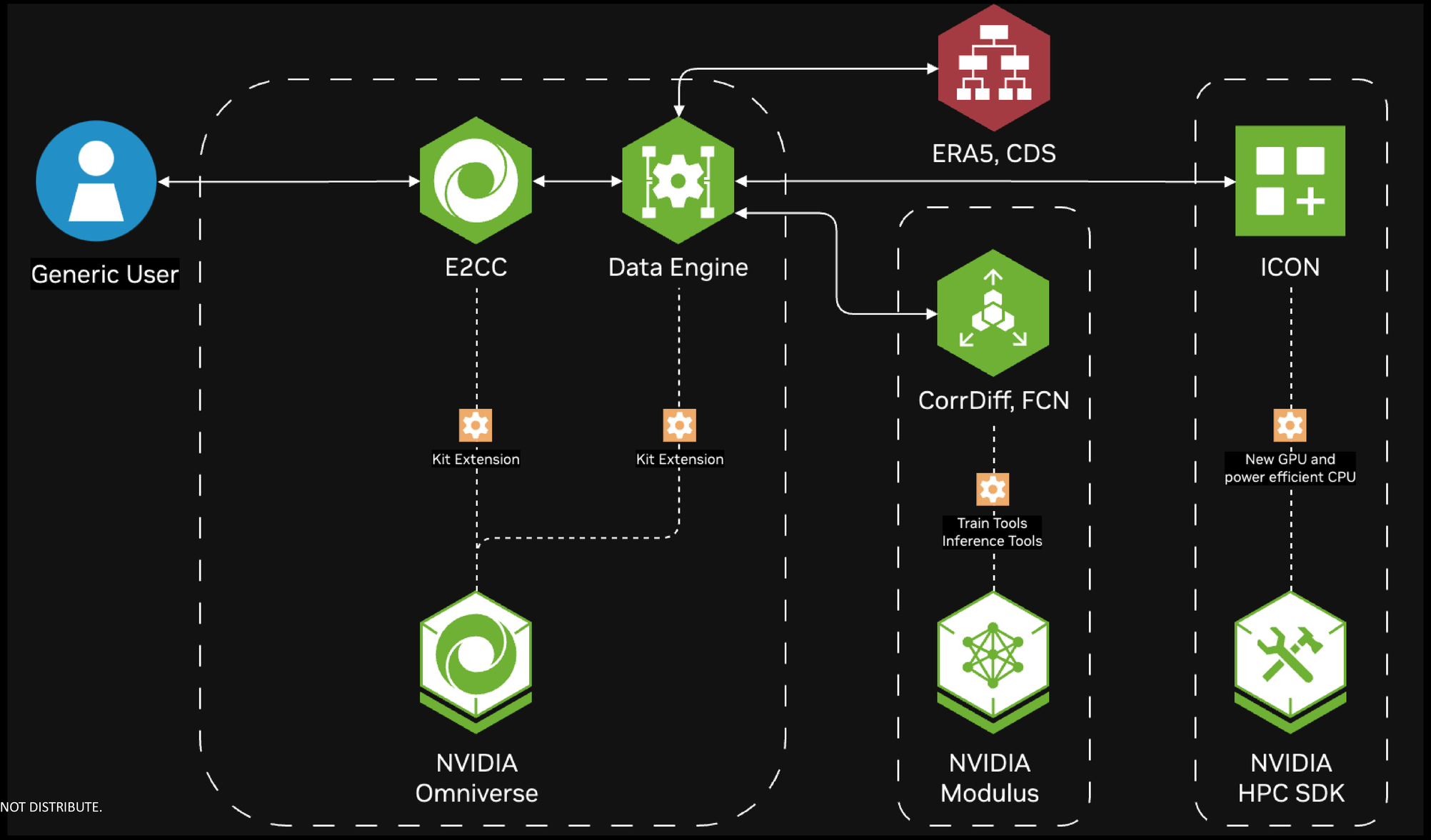
Earth-2 Goes Down to Street Level with City-Scale Simulation

VISUALIZATION

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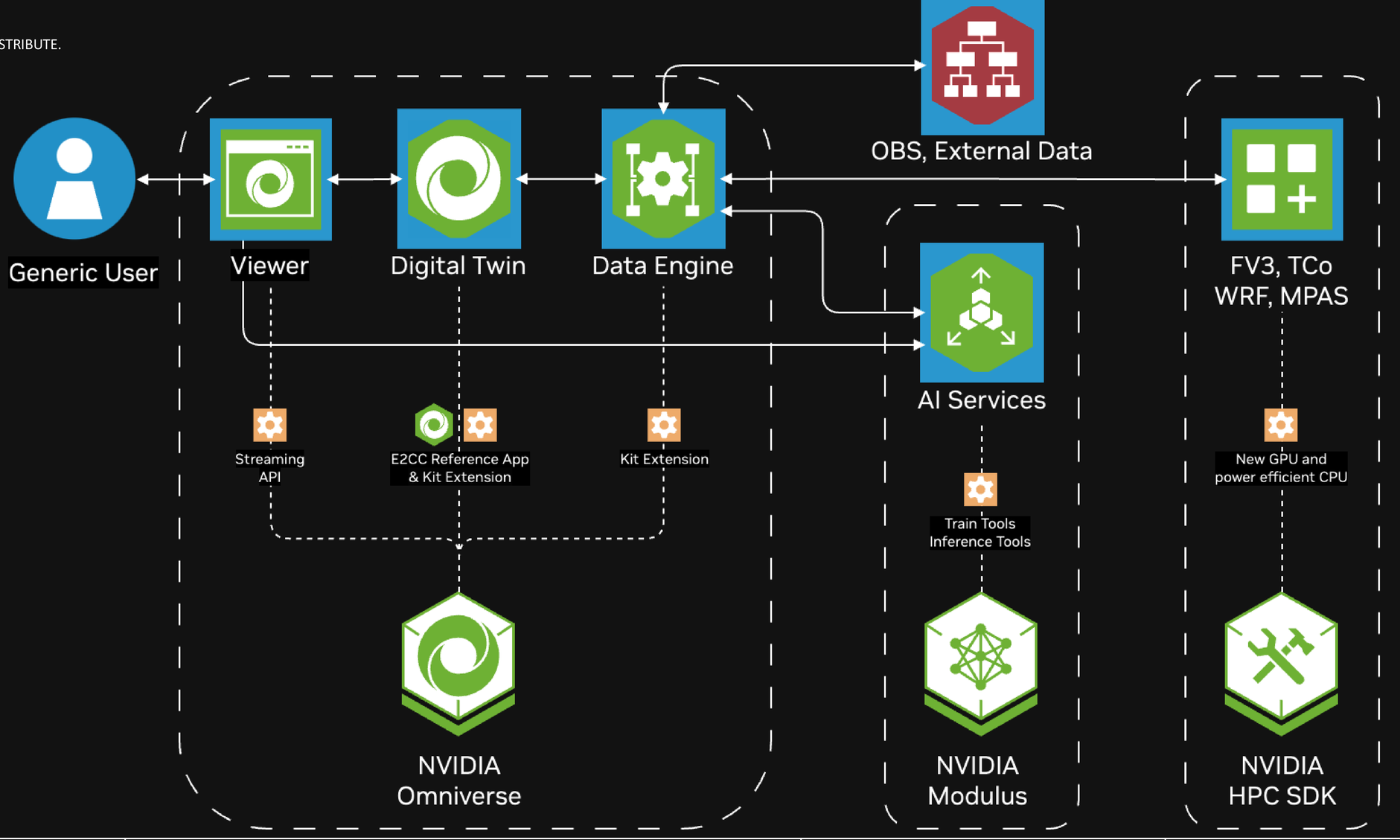
Earth-2 Service Architecture

Architecture demonstrated at GTC 2024



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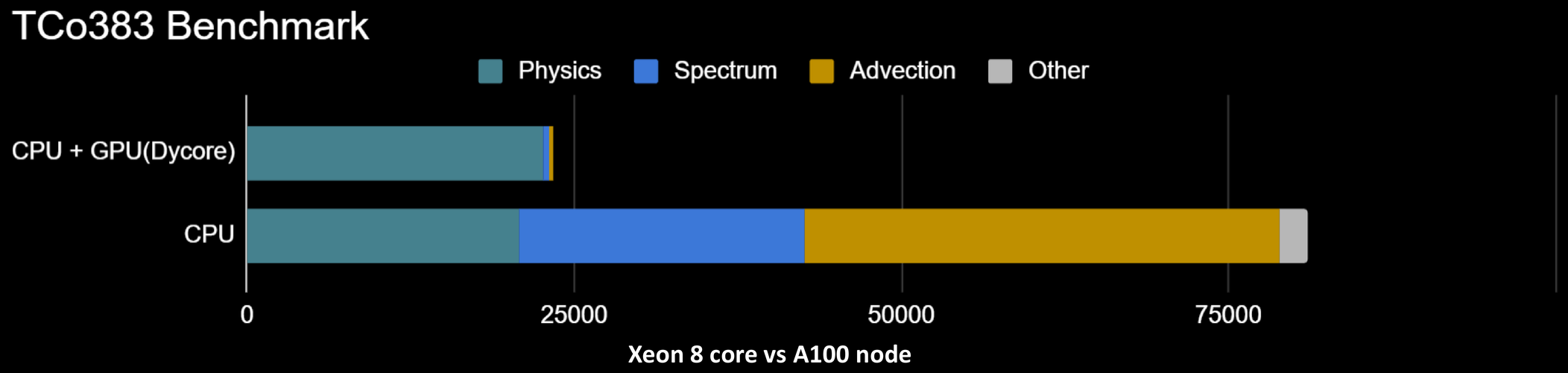
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	DIGITAL TWIN	AI	HPC
Use Cases	Streaming digital twin to developers/ users Rendering videos by OVX render farm	DLWPs (CorrDiff, StormCast … etc.) LLMs (TAIDE, NeMo … etc.)	NWPs (FV3, TCo, WRF, MPAS …) Data Processing, Mesh Generation, Verification …etc.
Personas	Front-end Engineer, Omniverse Engineer, Data Engineer	Data Scientist	Domain Scientist, Engineers
Comments	Recommendation: Virtual Machines with vGPU (vWS) - Workstation with RTX6000Ada - Server with L40S	CorrDiff A100 Ref Sizing - Inference: few seconds on 1 * A100 - Training: 7 Days on 128 * A100s	Status of Co-projects - [GPU - A100] TCo Dycore > 108X (node by node) - [CPU - Grace] WRF >12X (node by node)

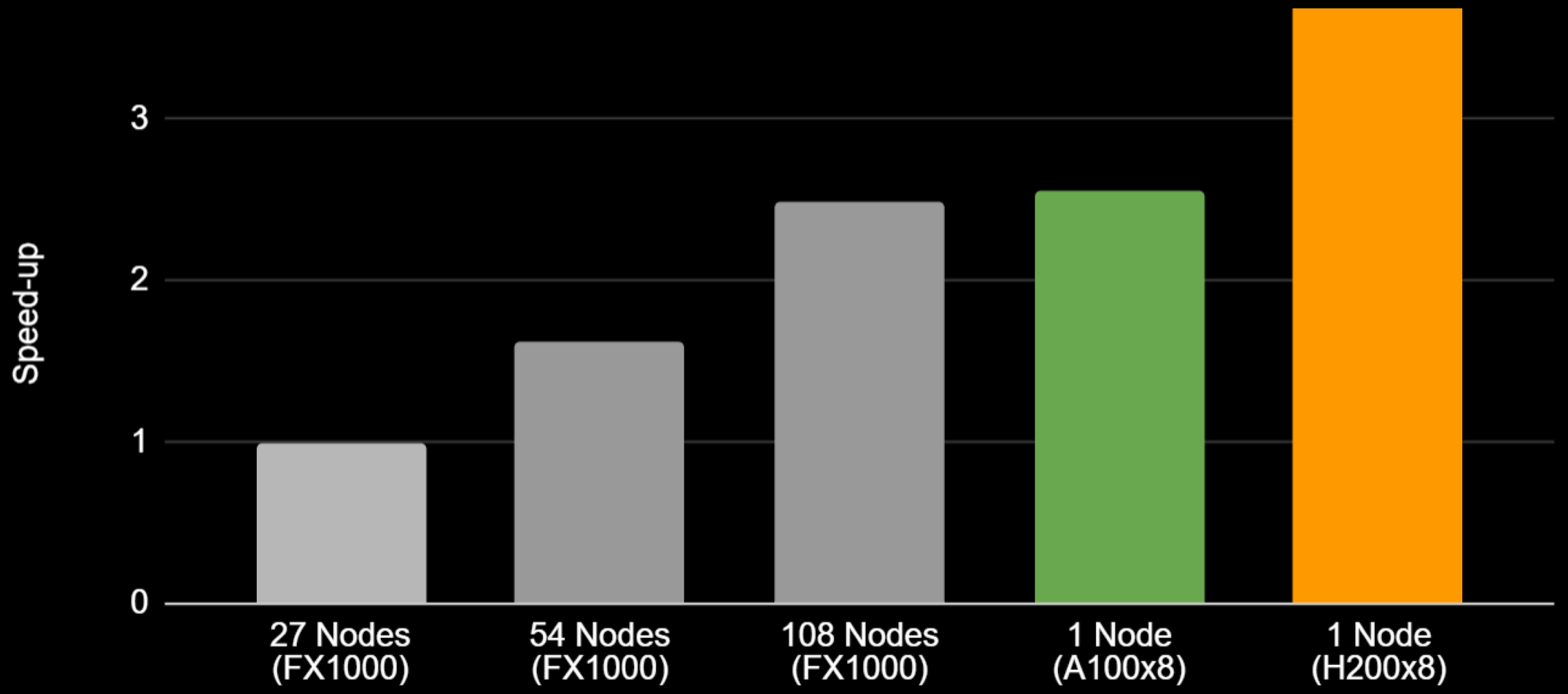
Current Status of TCo

- **Dynamic Core (Dycore)**:
  - The **Dycore**, which includes **advection** and **spectrum** operations, has been successfully **optimized** by porting its subroutines to **GPU**.
- **Physics Parameterization**:
  - Targeted optimizations for this component could bring similar performance improvements as seen with the dynamic core.



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CPU vs GPU for TCo639 Dycore



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WRF with Grace vs FX1000



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NVIDIA.

Summary

WRF (CPU)

6X+

Socket-by-Socket
Grace vs A64FX

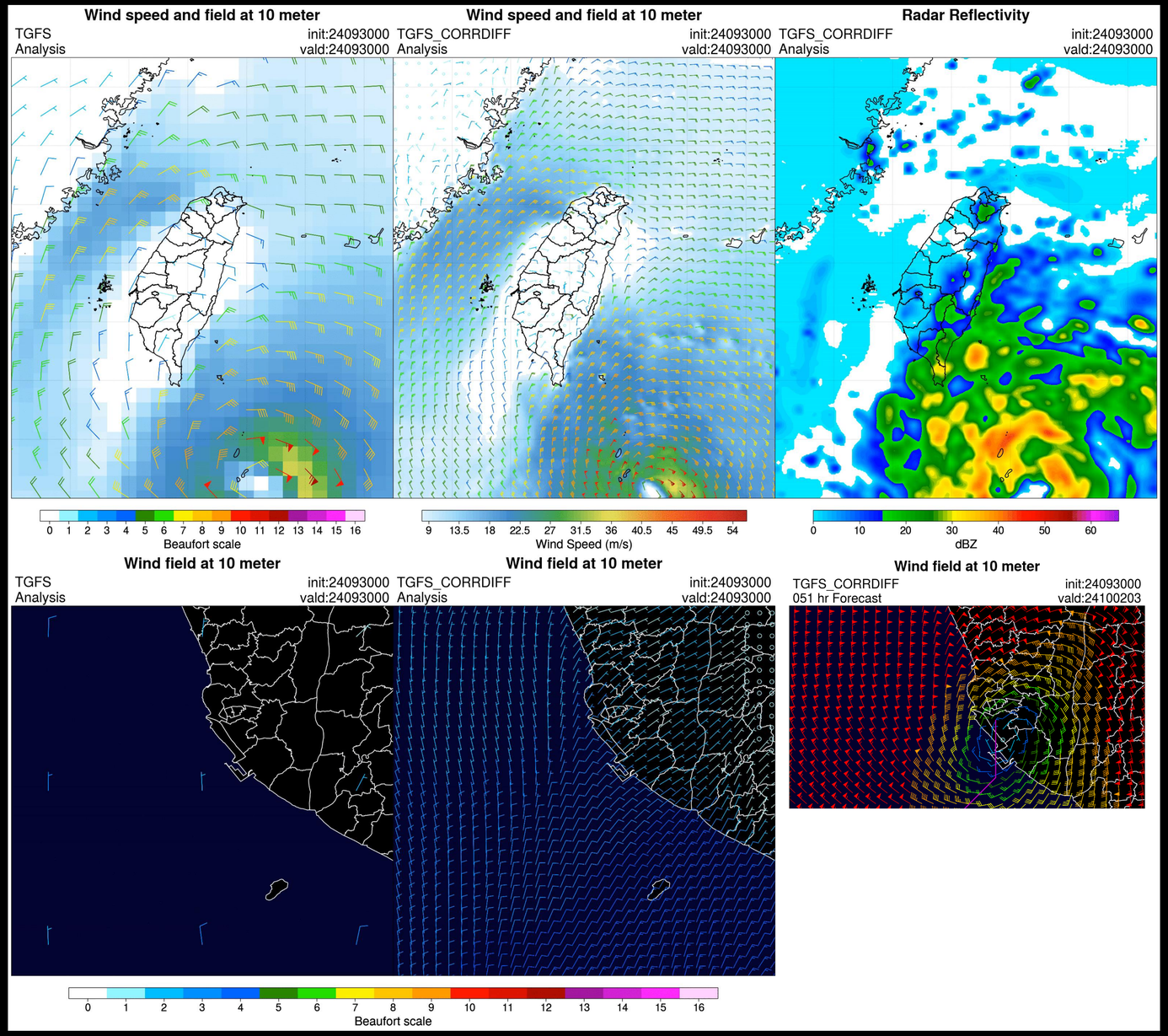
TCo (GPU)

108X+

Dycore, Node-by-Node
A100 vs A64FX

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CorrDiff as a Potencial Downscale Tools



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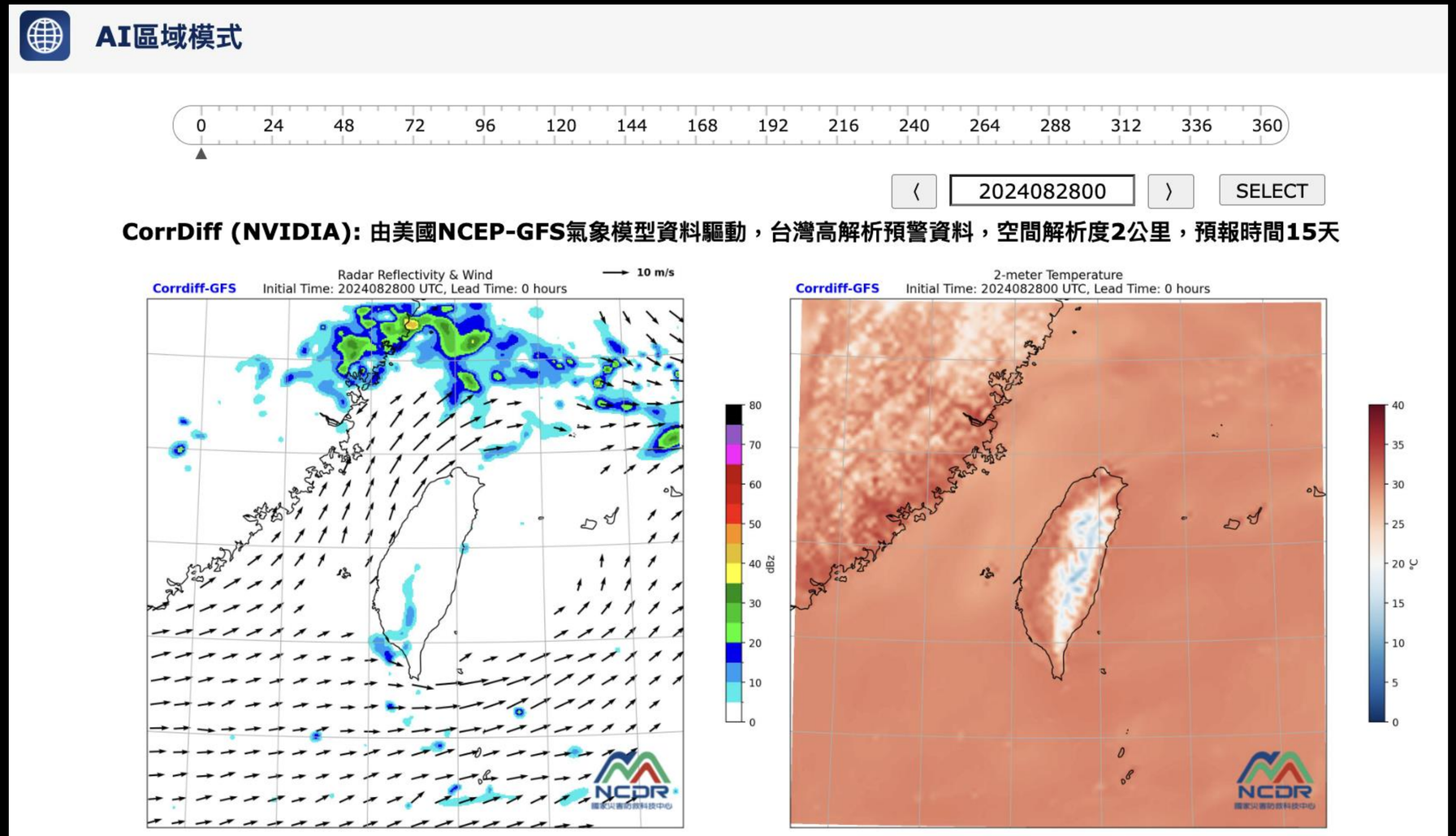
Creidit to: CWA: 劉正欽

NVIDIA

Aug 2024 - NCDR deploys CorrDiff on TWCC

Key Developers 江宙君, 朱容練, 于宜強 (NCDR) supported by **Jay Chen (NVIDIA)**

https://watch.ncdr.nat.gov.tw/watch_page_corrdiff

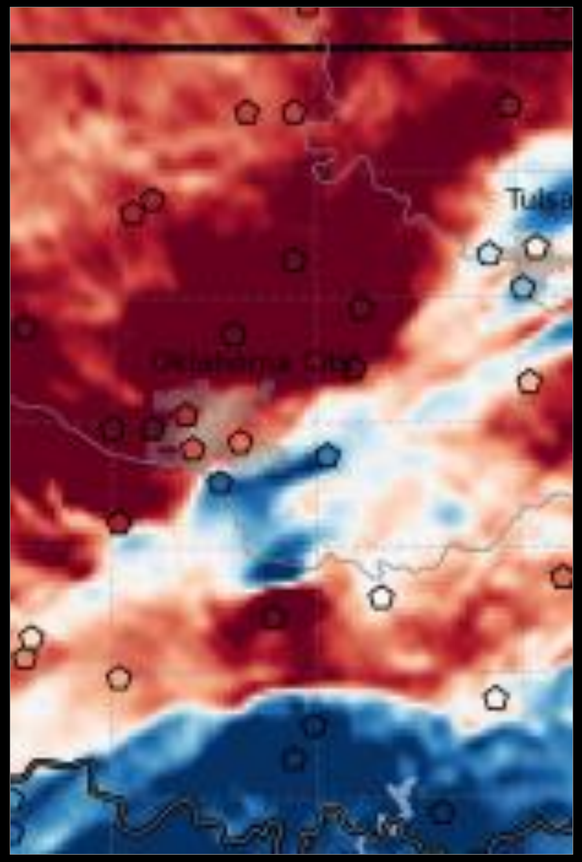


CorrDiff套件源自輝達公司(NVIDIA)，是利用ERA5與WRF資料所訓練出來的區域細緻化氣象模型
(https://catalog.ngc.nvidia.com/orgs/nvidia/teams/modulus/models/corrdiff_inference_package)

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The Latest Progress from NVIDIA Earth-2 Research Team

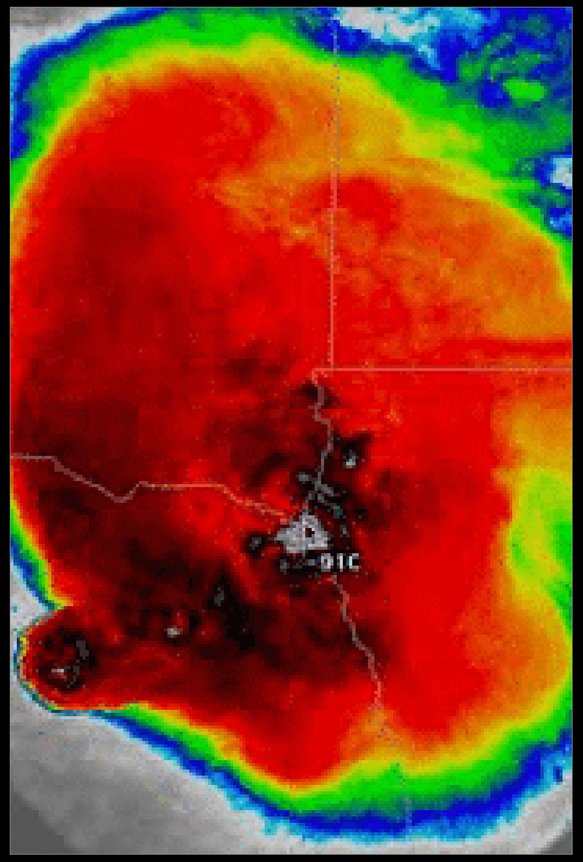
New techs will soon on Modulus and Earth2Studio.



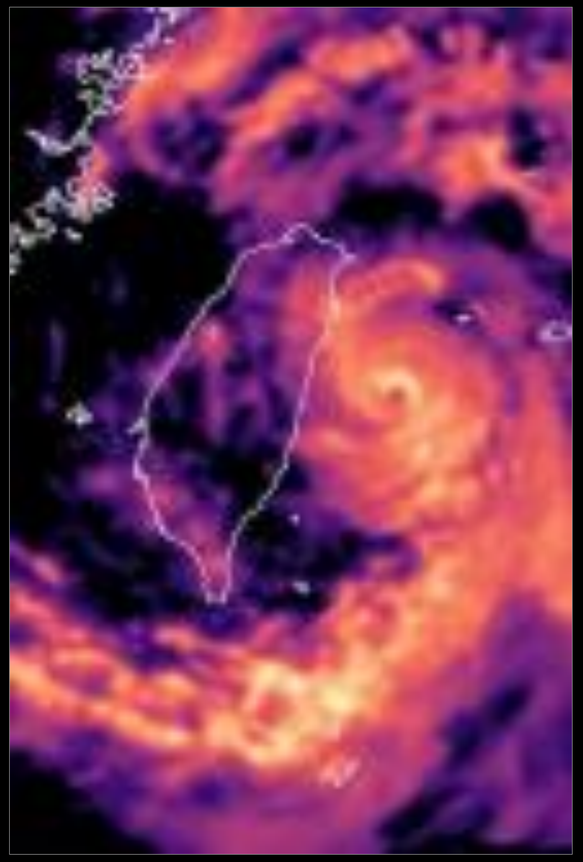
Stochastic Data Assimilation (SDA)
19 Jun 2024
https://arxiv.org/abs/2406.16947
Oxford, NVIDIA



Huge Ensembles (HENS)
6 Aug 2024
https://arxiv.org/pdf/2408.03100
https://arxiv.org/abs/2408.01581
LBNL, Berkeley, NVIDIA



StormCast
19 Aug 2024
https://arxiv.org/html/2408.10958v1
NVIDIA, LBNL, Washington

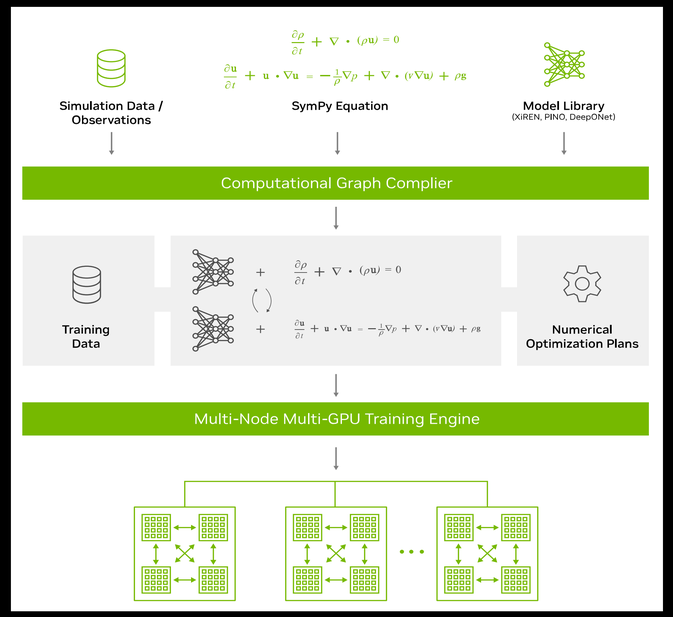


Stochastic Flow Matching (SFM)
28 Sept 2024
https://openreview.net/pdf?id=HZxJfzs3w6
UCI, NVIDIA

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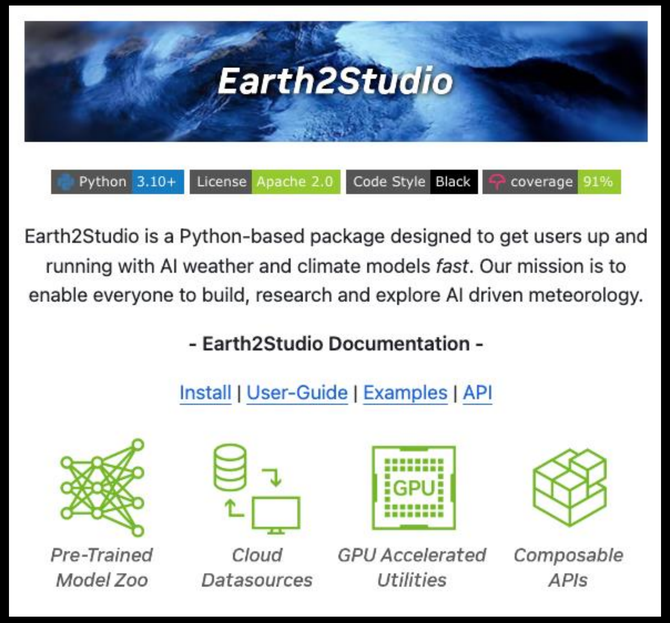
NVIDIA Earth-2 AI Tools

Modulus & Earth2Studio & NIMs



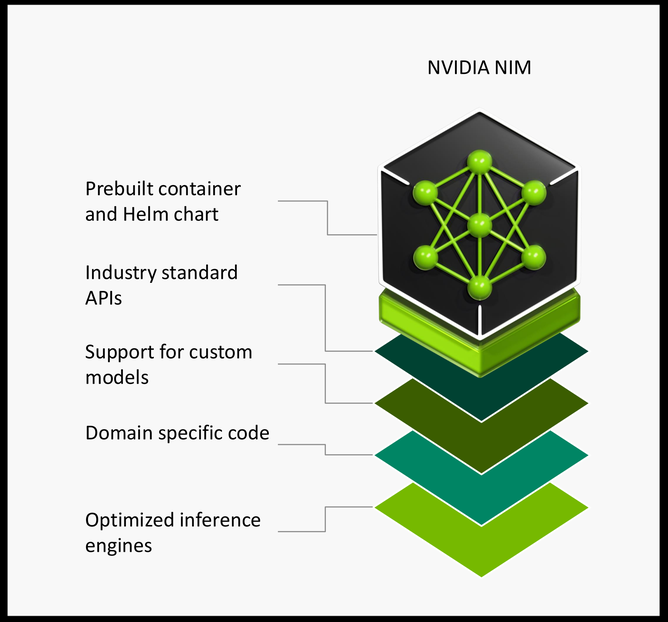
Modulus - Training

https://github.com/NVIDIA/modulus



Earth2Studio - Exploration

https://github.com/NVIDIA/earth2studio



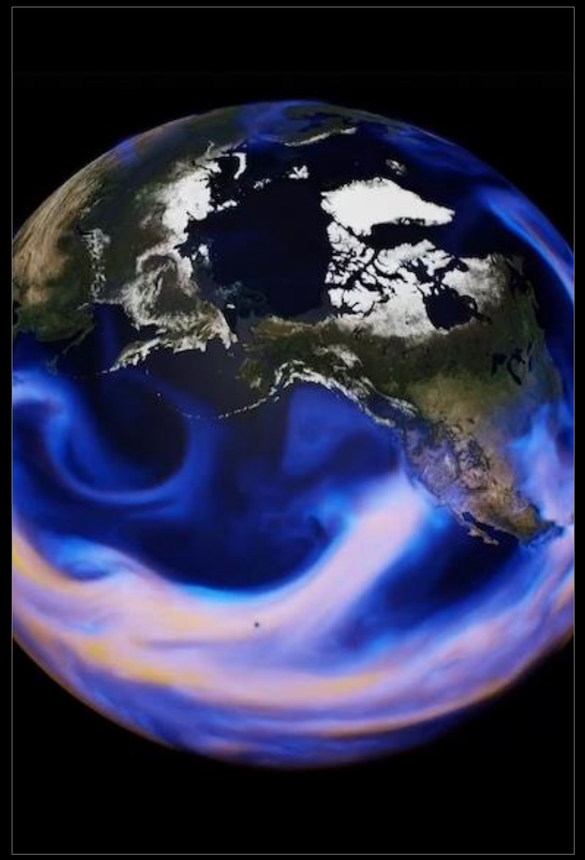
NIMs – Services

https://build.nvidia.com/nvidia/ai-weather-forecasting

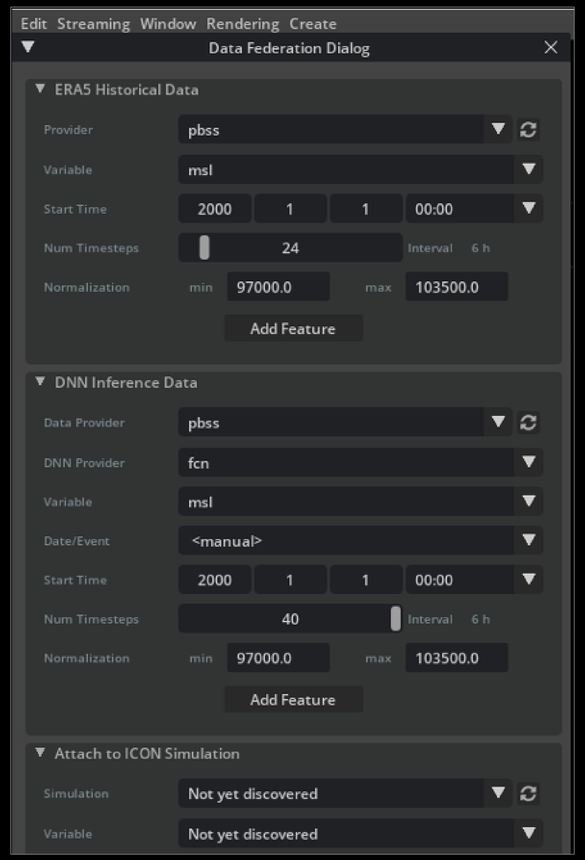
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CWA Earth-2 Visualization

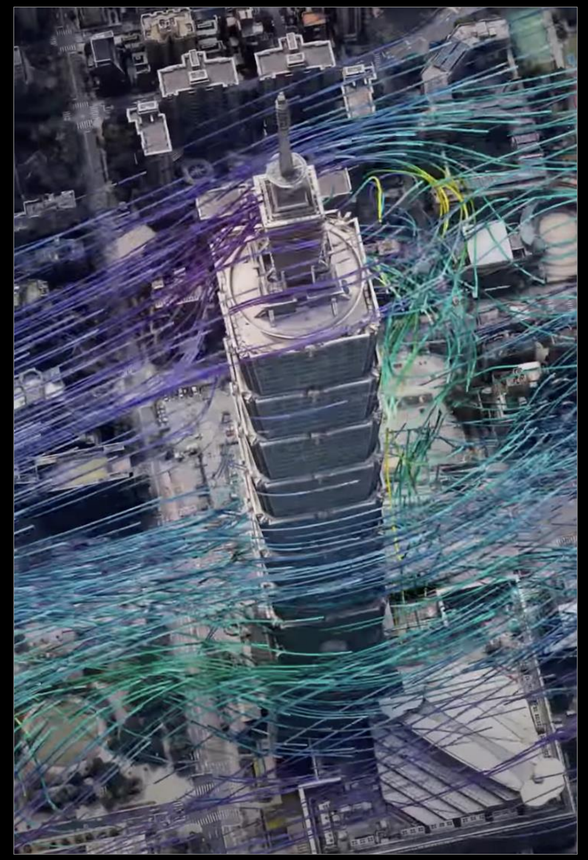
Recommended Stages



Visualizing Processed Image
Stage 1



Visualizing Multi-data Sources
Stage 2



Visualizing 3D and GIS Data
Stage 3



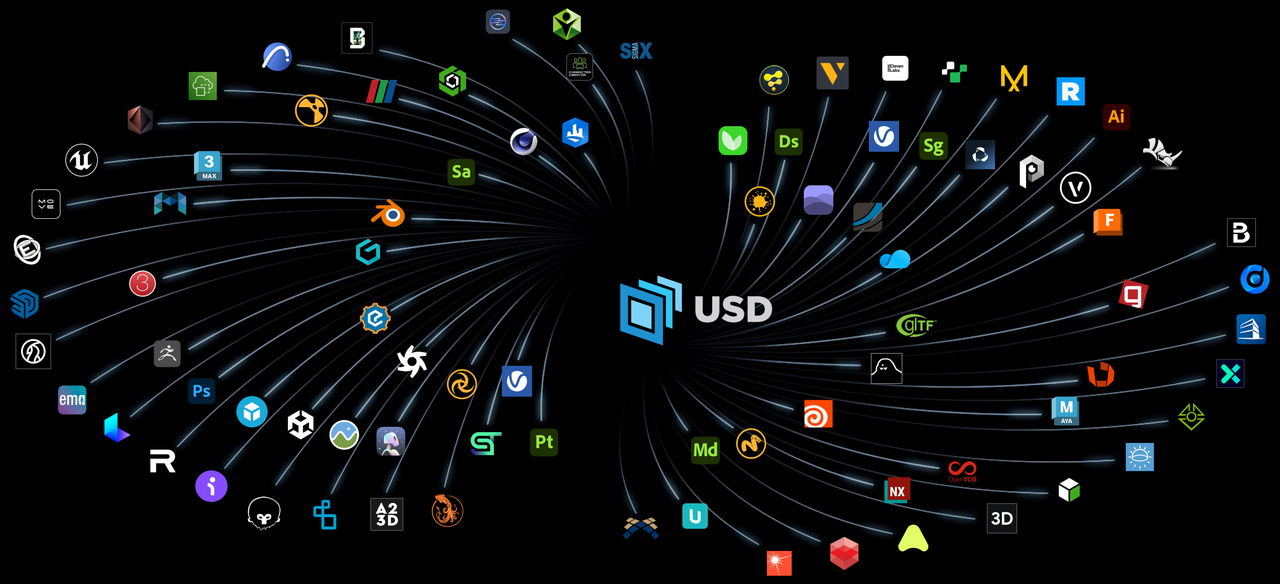
Connecting to Open Data
& DLWP APIs
Stage 4

Static Digital Twin

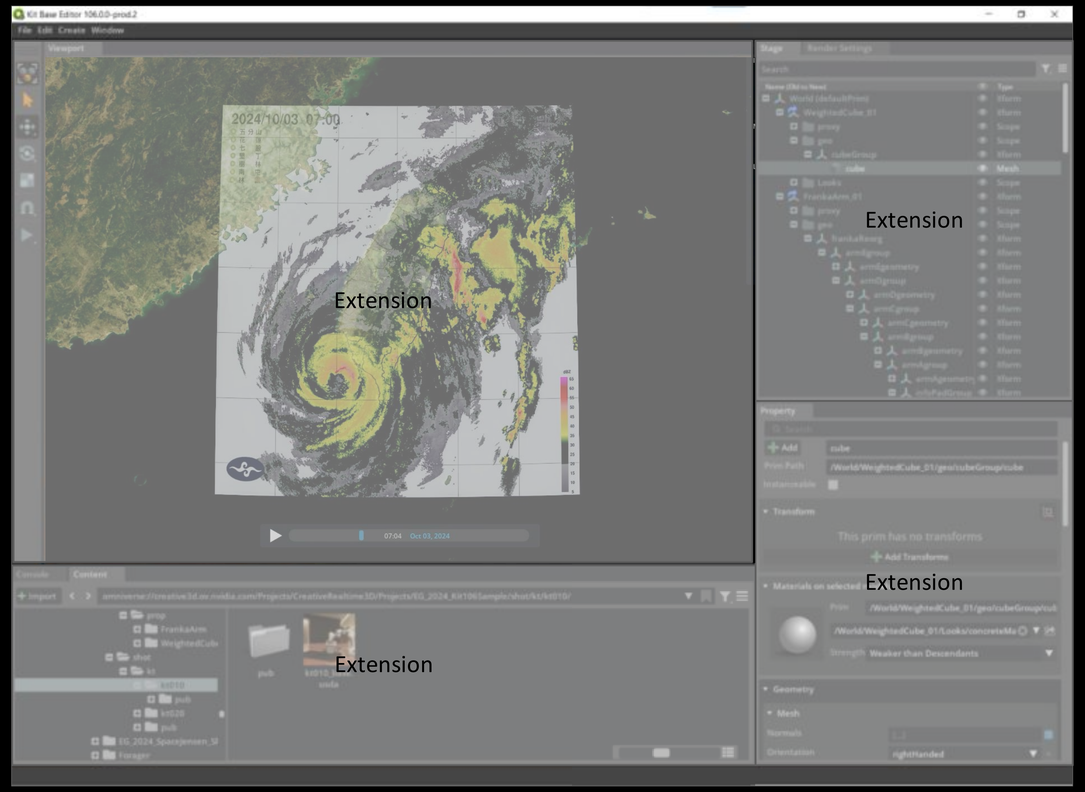
Dynamic Digital Twin

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Why NVIDIA Omniverse



OpenUSD Ecosystem Momentum



Scalability & Extensibility

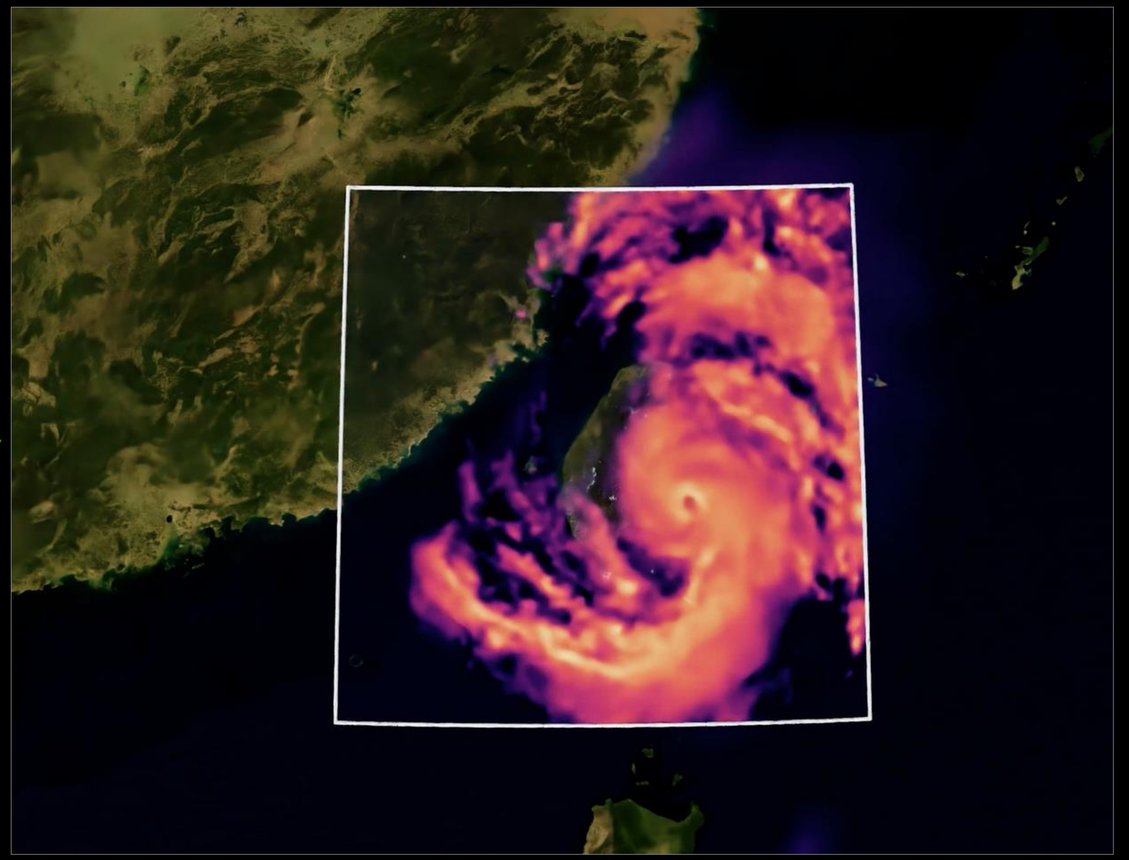
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Stage 1: Visualizing Processed Image

How CorrDiff was visualized by E2CC in GTC 24



Earth-2 Command Center (E2CC)

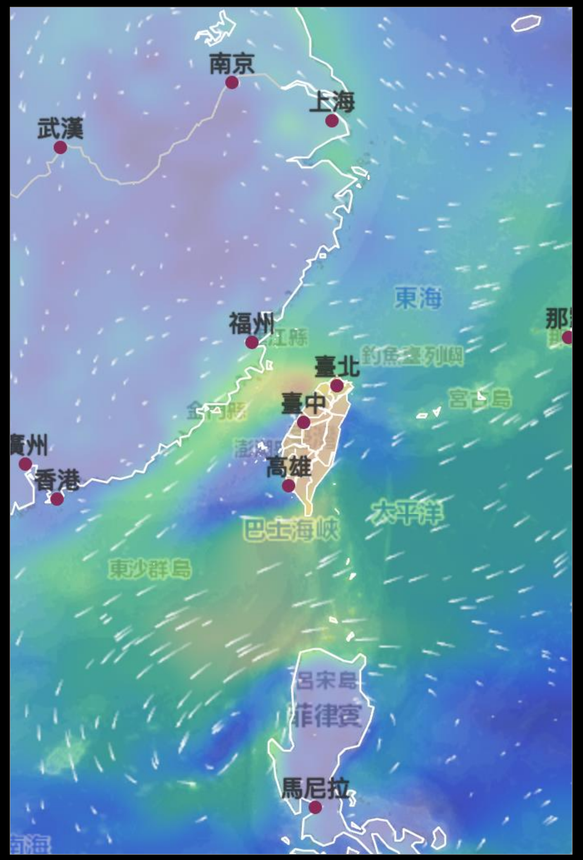


NVIIDA GTC 24 Earth-2 Demo

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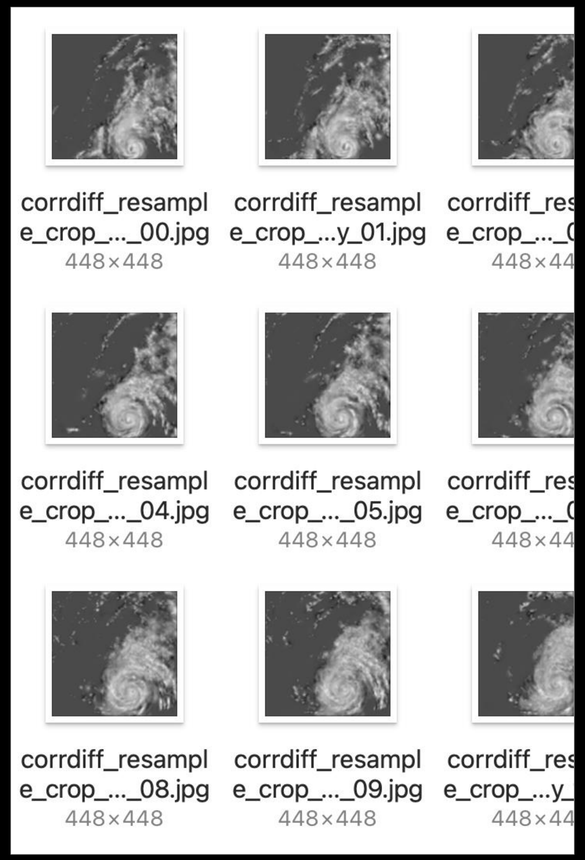
Stage 1: Visualizing Processed Image

How does it work?



Collect global NWP data

Step 1



Downscaled by CorrDiff
Save in JPG format

Step 2

```
def add_corr_diff_callback(ext):

    base_url = carb.settings.get_settings().get_as_string("
    path_pattern = base_url + '/corrdiff_test_001/corrdiff_

    # Create Timestamped Sequence
    seq = TimestampedSequence()
    idx = ext.get_next_idx()

    # create a unique url for this sequence
    import uuid
    seq.target_url = f'dynamic://test_sequence_{uuid.uuid1(

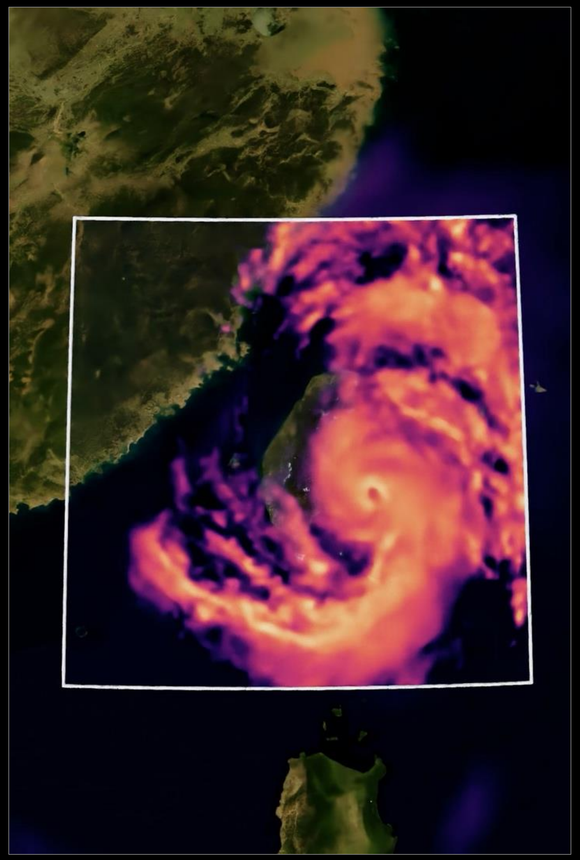
    import datetime
    start_time = datetime.datetime(year=1990, month=1, day=1
    # NOTE: That's how the timestamps were stored
    raw = [277824, 277825, 277826, 277827, 277828, 277829, 2
    timestamps = [start_time+datetime.timedelta(hours=v) for
    frame_skip = 1
    start_frame = 0
    end_frame = 38
    to_insert = []
    for i in range(start_frame, end_frame+1, frame_skip):
        cur_utc = timestamps[i]
        to_insert.append((cur_utc, path_pattern.format(frame
    seq.insert_multiple(to_insert)

    # create feature
    features_api = get_state().get_features_api()
    img = features_api.create_image_feature()
    img.sources = [seq.target_url]
    img.alpha_sources = img.sources

    from omni.earth_2_command_center.app.core.utils import T
    lon_min, lon_max = (116.39062500000000, 125.18225097656
    lat_min, lat_max = (19.613136291503906, 27.82678604125
    affine_mapping = latlong_rect_to_affine_mapping(
            lon_min, lon_max, lat_min, lat_max,
            is_in_radians=False)
    img.affine = list(affine_mapping.flatten()[0:6])
```

Load into E2CC
by a few lines of code

Step 3



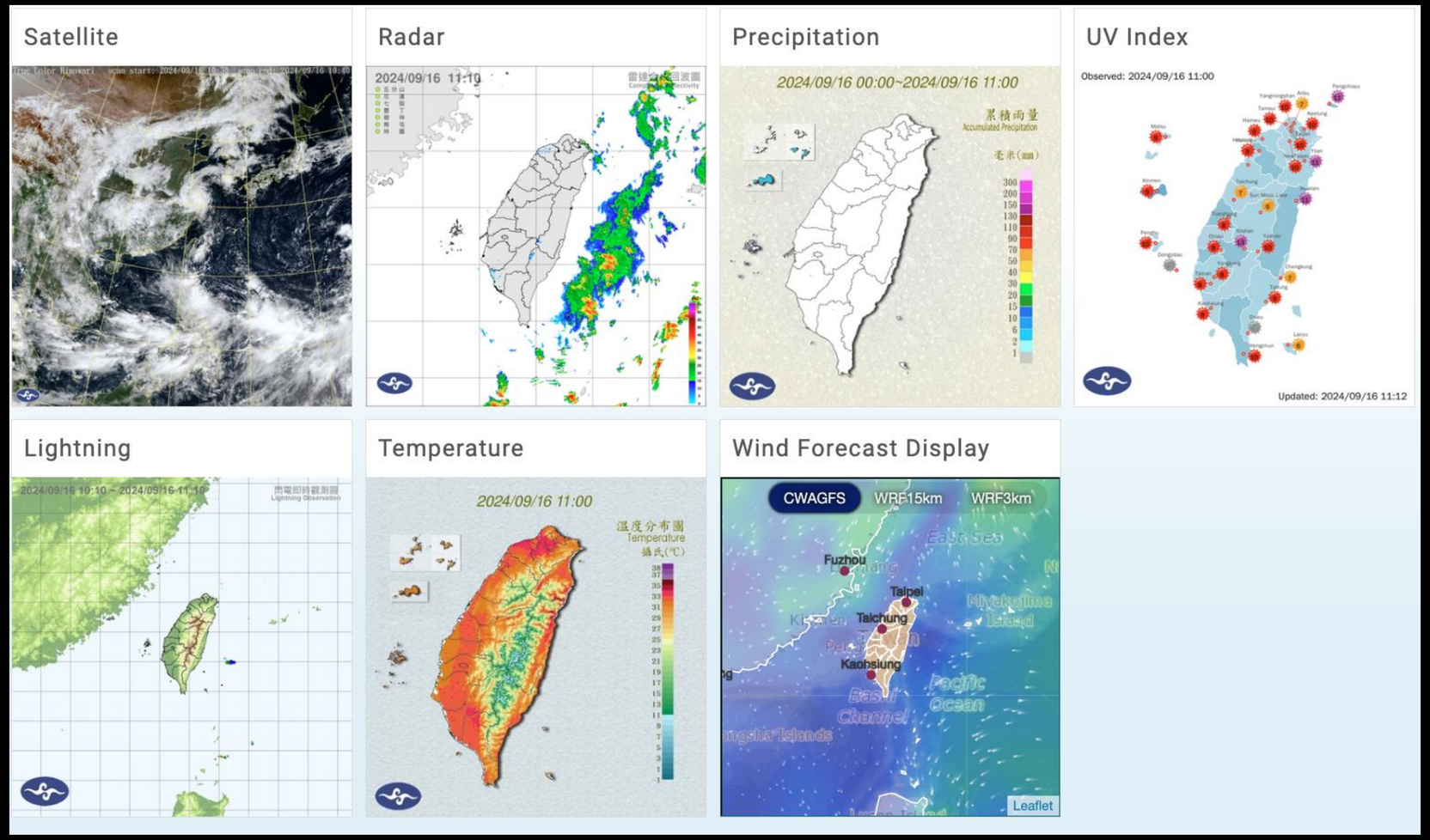
Show JPG data on E-2

Step 4

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Stage 1: Visualizing Processed Image

How to customize with CWA own dataset on E2CC?



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Earth-2 Command Center Dev 1.0.0 - New Stage*

Edit Window Rendering View Help

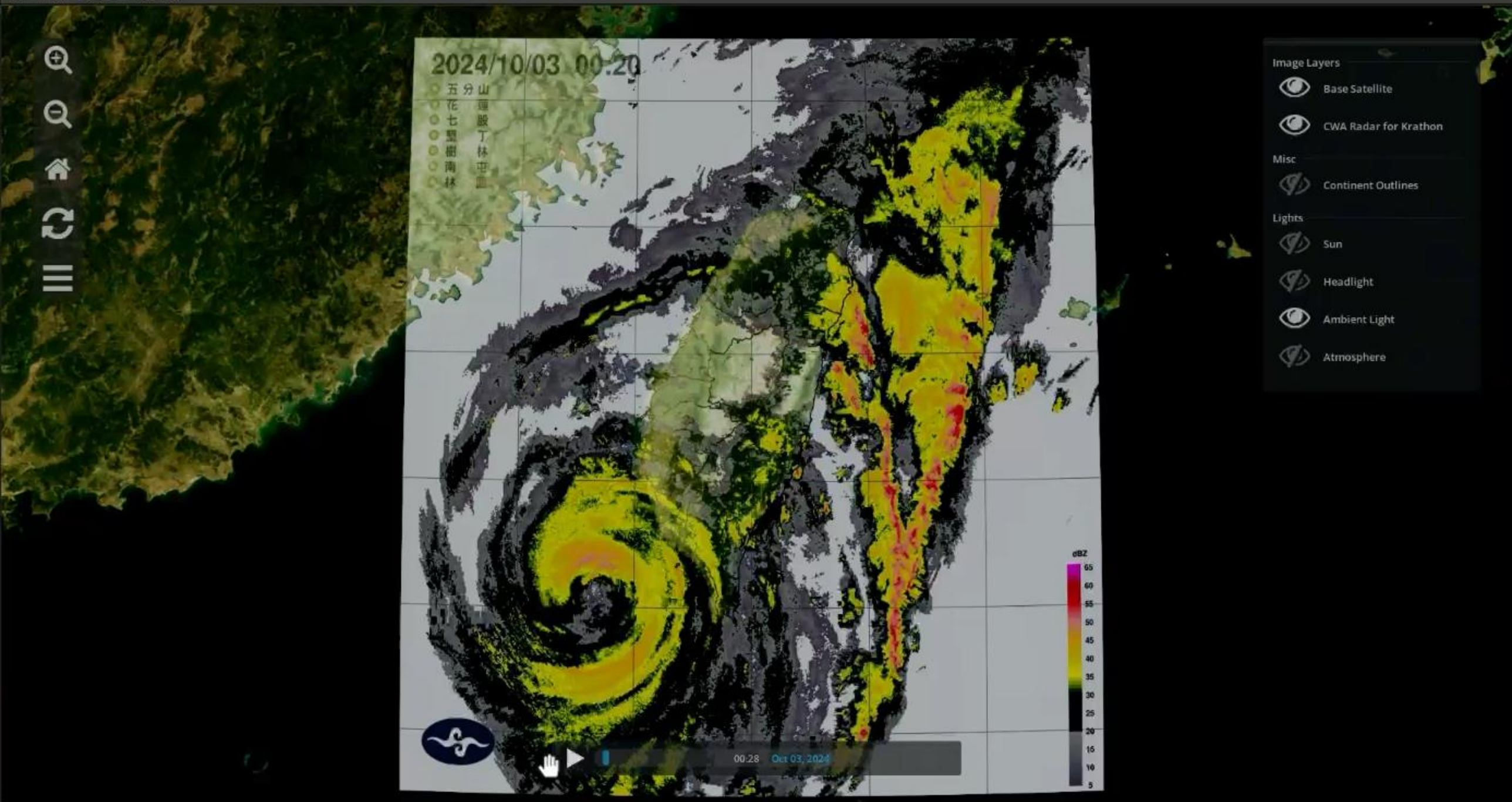


Image Layers

Base Satellite

CWA Radar for Krathon

Misc

Continent Outlines

Lights

Sun

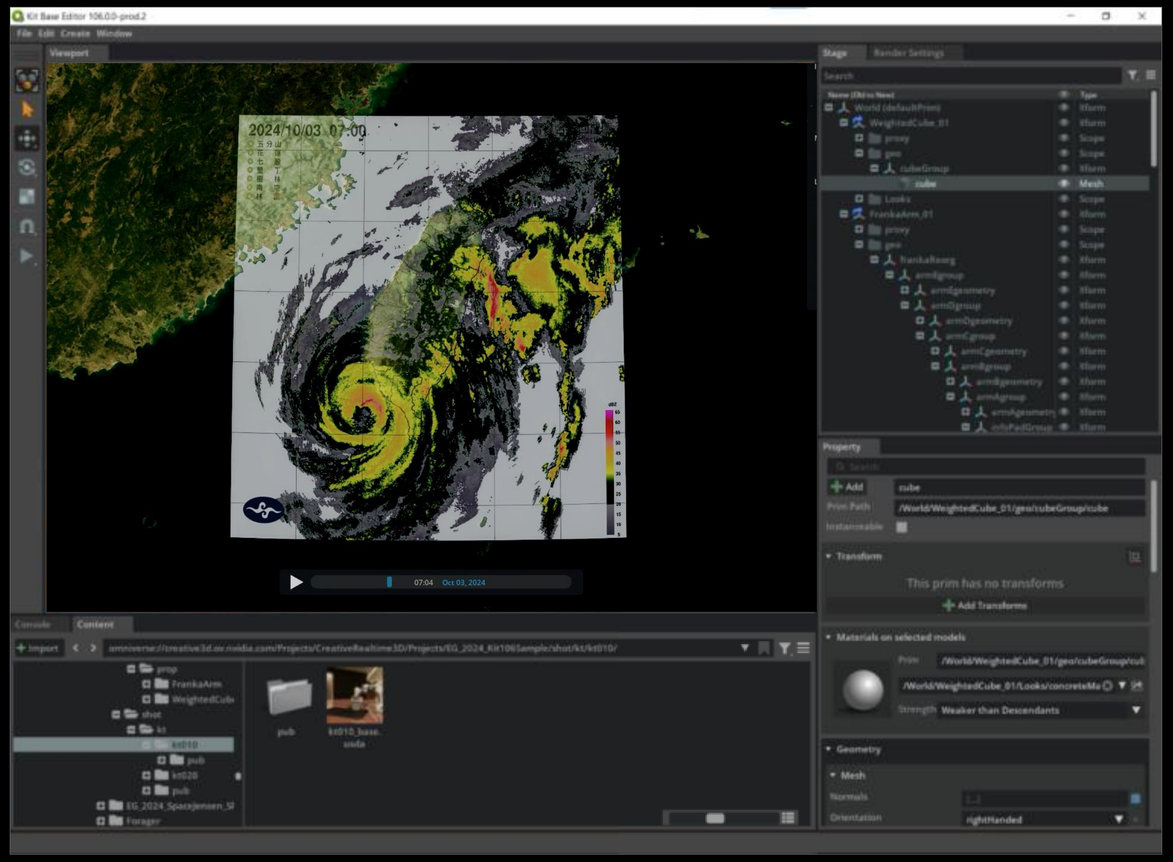
Headlight

Ambient Light

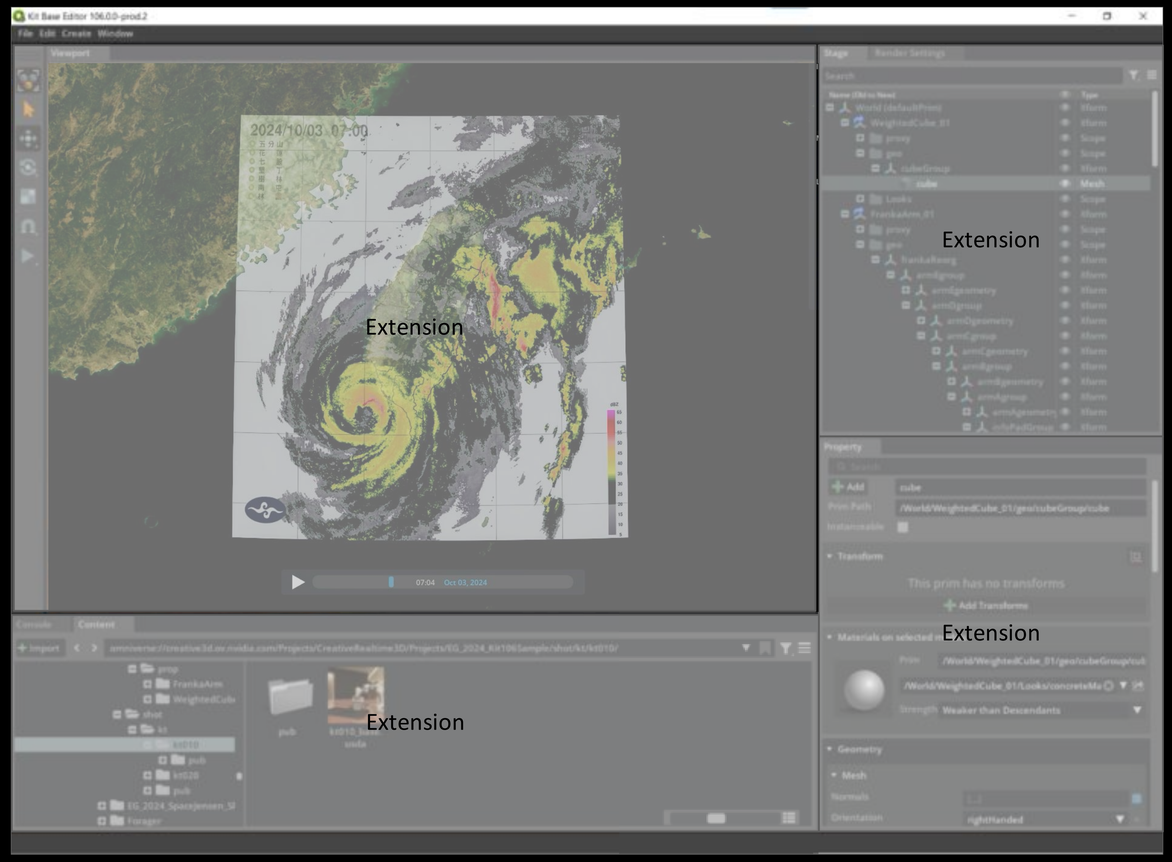
Atmosphere

Omniverse Kit-Based Applications

Modular, Extensible Development



Omniverse Kit Base Editor
Launch-ready Sample Environment



Omniverse Kit Extensions
Applications Composed of Modular Extensions

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A Concept DEMO: E2CC through OVAS



live broadcast at 1F lobby



embeded in website

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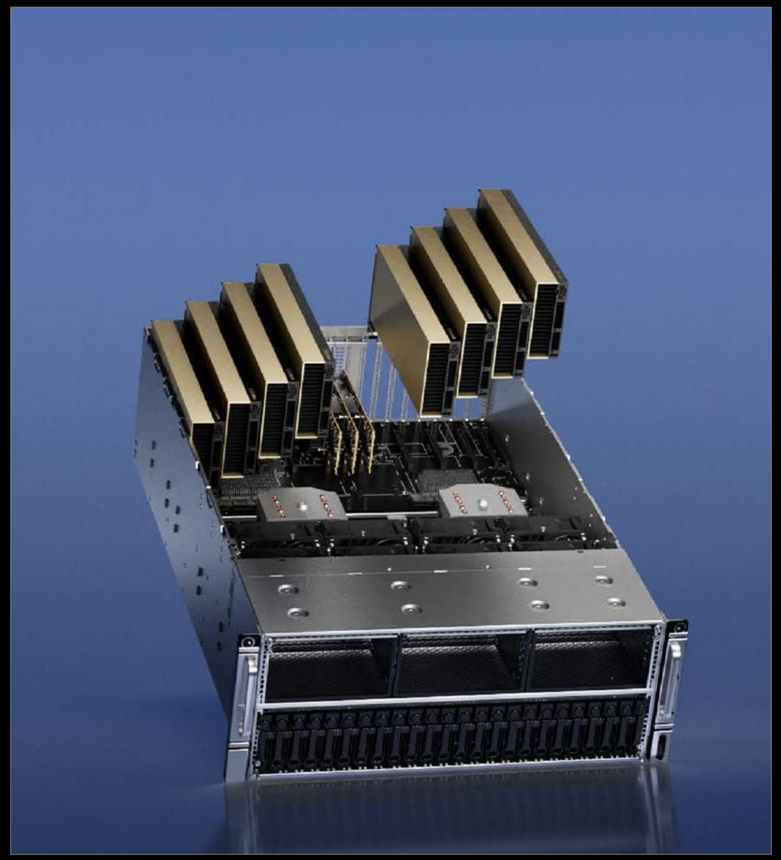
Omniverse Runs on NVIDIA RTX

NVIDIA-Certified Systems Optimized for Omniverse Workloads



RTX 工作站

Desktop: RTX 6000 Ada



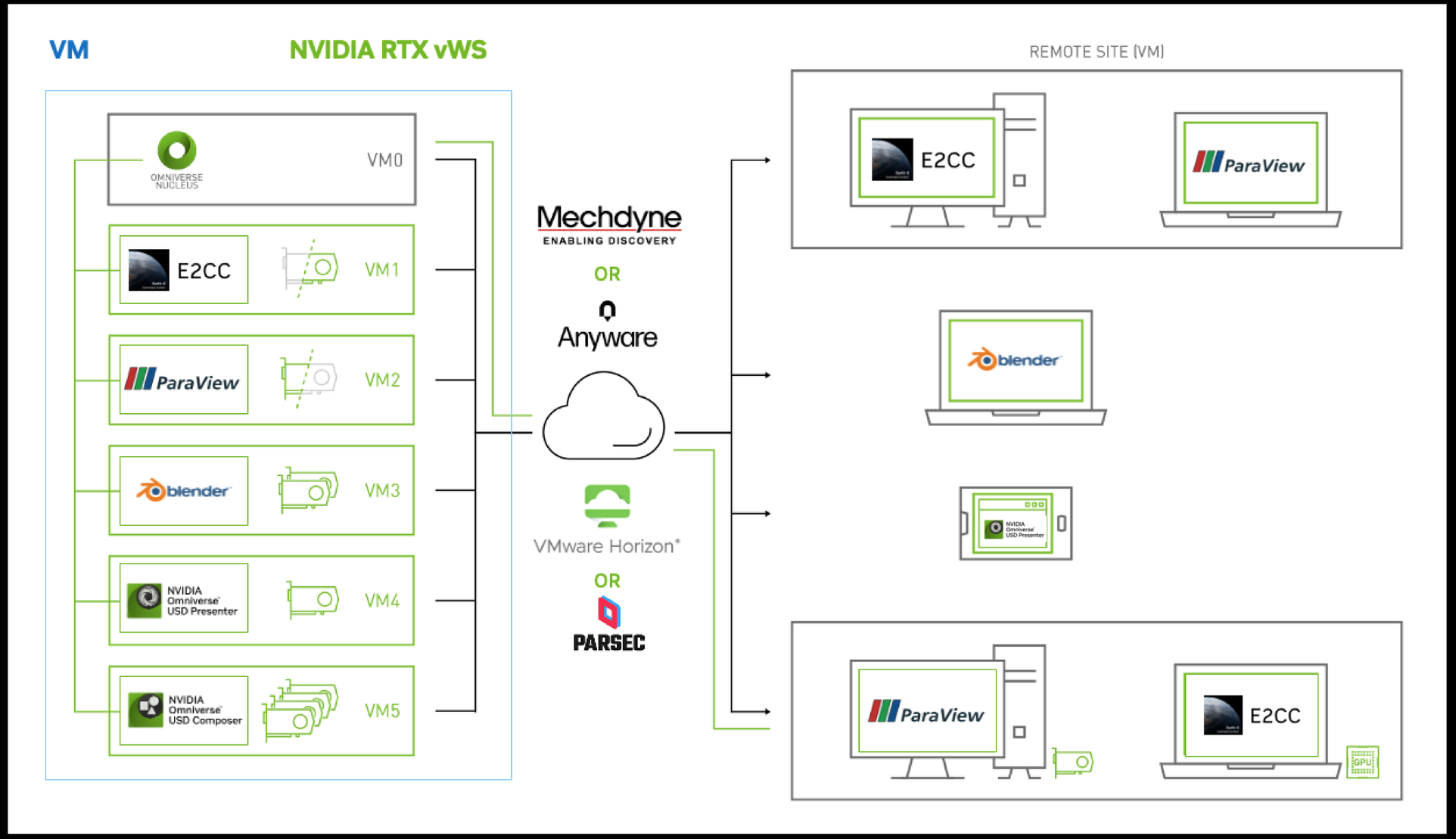
NVIDIA認證伺服器 (NCS)

L40S

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Reference Architecture

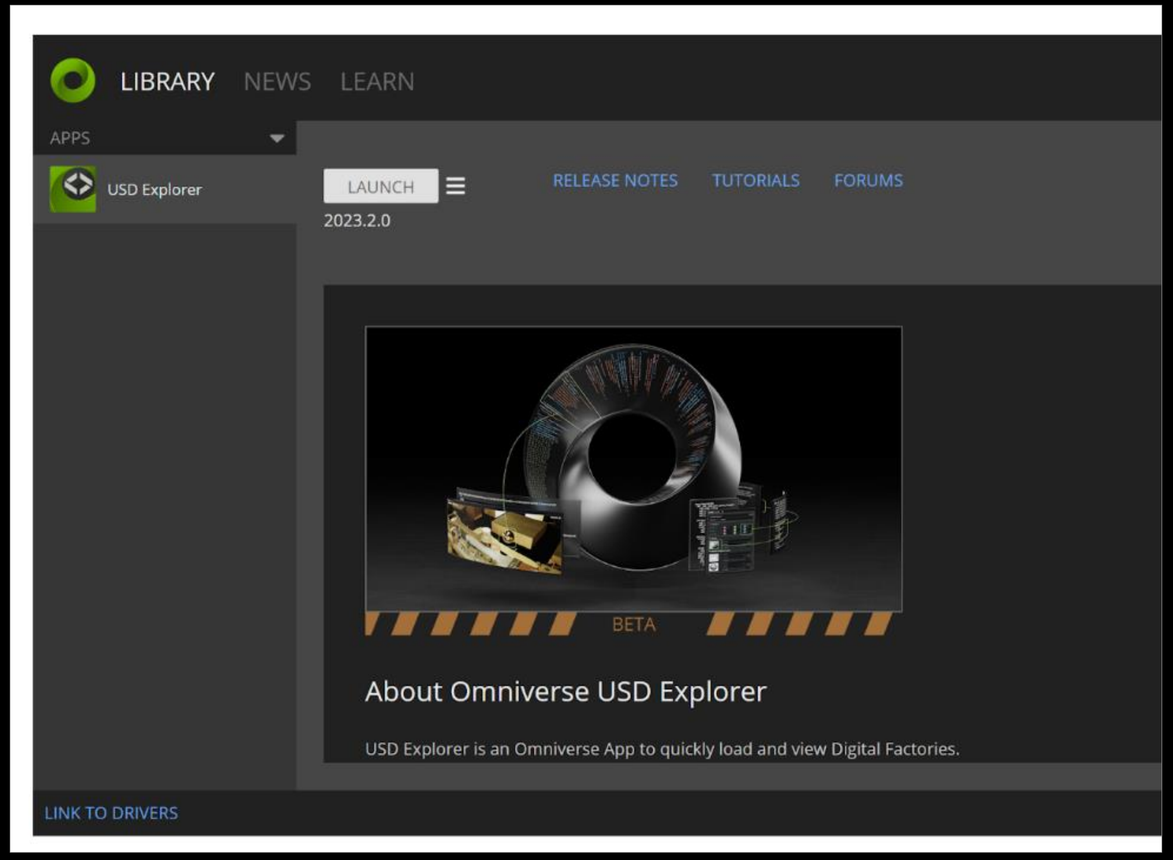
Virtualized Topology



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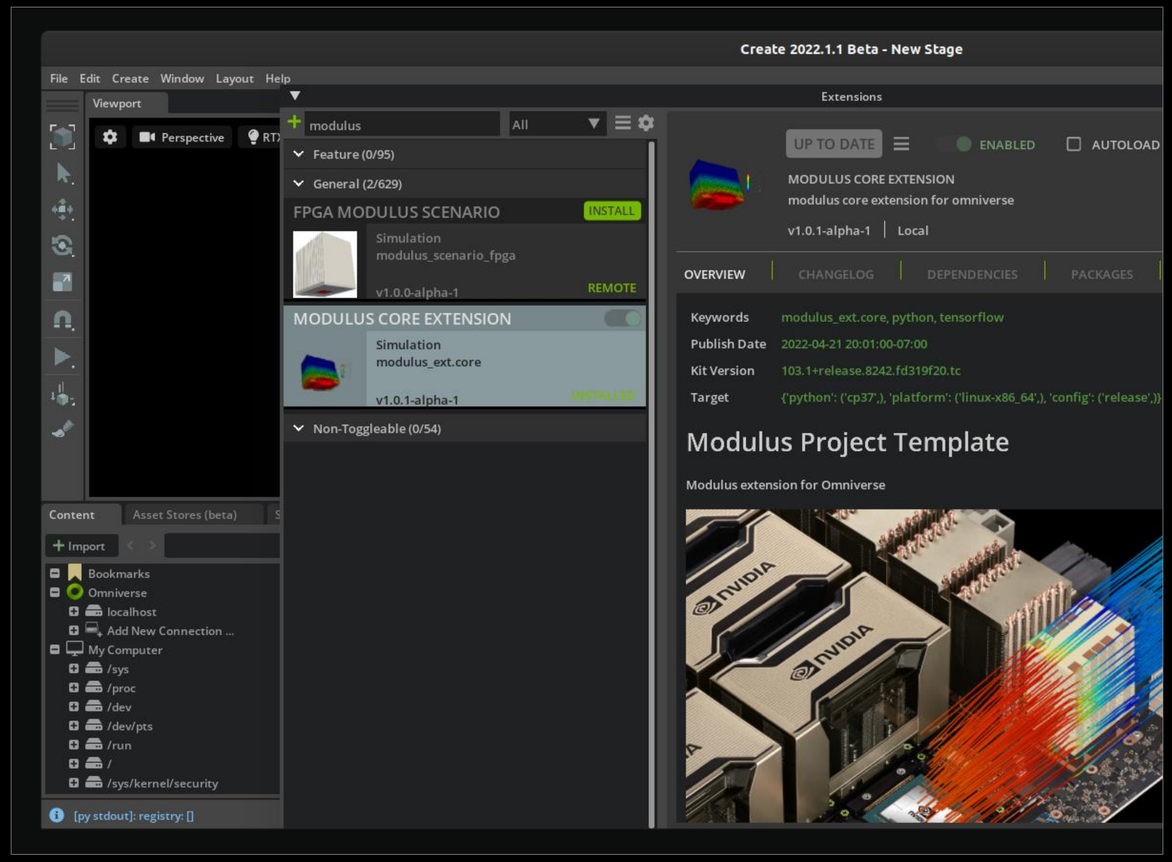
Omniverse Basics

Get started with the recommended courses



Developing an Omniverse Kit-Based Application (2 HR)

Course Detail | NVIDIA



Develop, Customize, and Publish in Omniverse With Extensions (8 HR)

Course Detail | NVIDIA

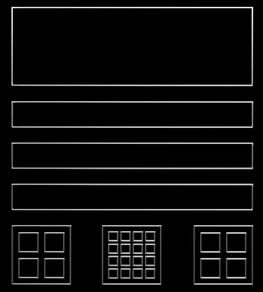
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Earth-2 @ CWA

WRF (CPU)

FV3 (CPU)

TCo (GPU)



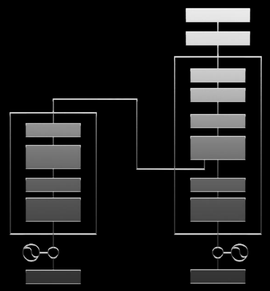
HPC SDK

HPC

FourCastNet-v2

CorrDiff

Coming Soon…



Modulus | Earth2Studio | NIM

AI

ParaView+IndeX

NVIDIA GTC 24

Computex 24



OMNIVERSE

VISUALIZATION

Earth-2 Platform

CPU | GPU | DPU

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Thank You

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